Improving External Validity of Machine Learning, Reduced Form, and Structural Macroeconomic Models using Panel Data

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We show that adding countries as a panel dimension to macroeconomic data can statistically significantly improve external validity of structural and reduced form models, as well as allow machine learning methods to outperform these and other macroeconomic forecasting models. Using GDP forecasts for evaluation, this procedure statistically significantly improves root mean squared error (RMSE) of 12% across horizons and models for a selection of reduced form models. We also show statistically significant RMSE improvements of 24% across horizons for structural DSGE models. Removing US data from the training set and forecasting out-of-sample country-wise, we show that we can make both reduced form and structural models more policy invariant, and even without US data in the training set, we can statistically outperform on the US GDP test set compared to a baseline model that uses US data only. Finally, given the comparative advantage of "nonparametric" machine learning forecasting models in a data rich regime, we demonstrate that our recurrent neural network (RNN) model and automated machine learning (AutoML) approach outperforms all baseline economic models in this regime. Furthermore, robustness checks indicate that machine learning outperformance is reproducible, numerically stable, and generalizes across models.

JEL Codes: E27, C45, C53, C32
Keywords: Neural Networks, Policy Invariance, GDP Forecasting, Panel Data

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I. Introduction

A central problem in both reduced form and structural macroeconomics is the dearth of underlying data. For example, GDP is a quarterly dataset that extends back only to around the late 1940s, which results in around 300 time steps. Thus generalization and external validity of these models are a constant problem. In forecasting, this approach is partially addressed by using simple linear models. In structural macroeconomics, this issue also does not have a satisfactory resolution, but the use of micro-founded parameters and Bayesian estimation attempts to improve generalization to a limited effect.

The main contribution of our paper is to propose augmenting single country data with a panel of other countries to improve external validity. Unlike factor models (Stock and Watson (2002*a*) and Stock and Watson (2002*b*)), this does not increase the number of covariates, but increases the number of data points. For example, when forecasting US GDP, our model would use French macroeconomic data as well as many other countries to augment the number of observations/time steps in our training set.

Estimating traditional reduced form models–AR(2) (Walker, 1931), VAR(1), VAR(4) models (Sims, 1980)–we show that we can statistically significantly improve root mean squared error (RMSE) performance by an average of 12% across horizons and models. Outside of pure forecasting, we make reduced form models more policy/regime invariant. To show this, we estimate these models on data from all countries except the target country being forecasted. Thus, our test data is both time step out-of-sample, as is usual, but also country out-of-sample, which is not. Across most models and forecasting horizons, our out-of-sample forecasts outperforms the typical procedure of estimation models on only the data of the country of interest. Our augmentation leads to about 80% of the improvement gained from moving all the way to forecasting with the full data

set including the in-sample country. We think this provides suggestive evidence that this data augmentation scheme can help make reduced form models more policy invariant.

We proceed to apply our panel data augmentation scheme to structural DSGE models as well. For the sake of simplicity, we test our panel data approach on the common Smets-Wouters model (Smets and Wouters, 2007). Our augmentation statistically significantly improves the external validity of this model by an average of 24% across horizons. We again test our model out-of-sample, and show improvements over single country baselines. This suggests even DSGE models are not immune to the Lucas critique and the use of country panel data can improve policy/country invariance. Given the consistent improvements across all horizons and across reduced form models, we are confident this approach will generalize to estimation of other structural models. We would also advocate applying this approach to calibration, GMM (Hansen and Singleton, 1982), and Bayesian Estimation (Fernández-Villaverde and Rubio-Ramírez, 2007), where the macro-economist should attempt to match moments on a broad cross section of countries instead of just the United States, for example.

Recognizing that this augmentation increases the effective number of observations by a factor of 10, we also demonstrate that "nonparametric" machine learning models can now outperform traditional forecasting models in this high data regime. We use two different algorithms. The first algorithm is the use of automated machine learning (AutoML). This algorithm runs a horse race of hundreds of machine learning models to determine the best performing model on the validation set, which is ultimately tested on the test set. We view AutoML as a proxy for great machine learning performance, but imagine individual model tuning can improve performance even more. As different models perform

the better under the low data (US) regime and the high data (panel) regime under AutoML, we also test a recurrent neural network to show the improvement of a single model under both data regimes. The model improvement indicates that while these approaches are competitive in the low data regime, machine learning methods consistently outperforms baseline economic models–VAR(1), VAR(4), AR(2), Smets-Wouters, and factor models–in the high data regime. Furthermore, while our model uses a cross section of countries data, we only use three covariates–GDP, consumption, and unemployment lags. In contrast, the DSGE model uses 11 different covariates and the factor model uses 248, suggesting our machine learning models still have room for improvement. Over most horizons, our model approaches Survey of Professional Forecasters' (None, 1968) median forecast performance, albeit evaluated on 2020 vintage data (see Appendix C), resulting in outperformance over SPF benchmark at 5 quarters ahead. Moreover, the outperformance of our model over the SPF benchmark is noteworthy as the SPF is an ensemble of both models and human decision makers, and many recessions like the recent COVID recession are not generally predicted by models.

The paper proceeds as follows: Section II. reviews the literature on forecasting and recurrent neural networks and describes how our paper merges these two fields; Section III. discusses feed-forward neural networks, linear state space models, and gated recurrent units (Cho et al., 2014); Section V. describes the data; Section III.B. briefly mentions our model architecture; Section IV. discusses the benchmark economic models and the SPF that we compare our model to; Section VI. and Appendix G.3 provide the main results and robustness checks; and Section VII. concludes the paper.

II. Literature Review

This paper connects multiple strands of literature: machine learning, time-series econometrics, and panel macroeconomic analysis:

We apply a recently machine learning technique to our economic data–automated machine learning or AutoML. AutoML was a technique introduced and improved upon as machine learning models became more complicated with more layers and hyperparameters to tune (Thornton et al., 2013). Unlike deep learning, many innovations came from the software industry to automate estimation techniques. However, there is a vibrant academic literature following its introduction in Hutter F and H. (2014). The basic premise is to automate the model training and model discovery portion of machine learning. H2O (LeDell and Poirier, 2020), the AutoML platform we use, takes data and automatically runs a horse race of machine learning models on the validation set and returns the best performing model. Thus, we view the output model as a proxy for a well trained and effective predictive model trained by a data scientist, even if some additional fine tuning can improve performance. Although there is room for improvement, the automation of the entire model estimation pipeline, removing the human from the loop, shields us from typical p-hacking critiques.

The second machine learning procedure we used was the estimation of a recurrent neural network (RNN). A recurrent neural network is a state space model much like linear state space models used in economics. However, innovations in deep learning have improved the predictive power of these models over what economists are used to for their linear counterparts. RNNs have been around in many forms, but were mainly popularized in the 1980s (Rumelhart, Hinton and Williams, 1986). The popularity and performance of recurrent neural networks grew with the introduction of long short-term memory (LSTM) networks with Hochreiter and Schmidhuber (1997). The model uses gates to

prevent unbounded or vanishing gradients giving this model the ability to have states that can "remember" many time steps into the past. In addition, to its pervasive use in language modeling, long short-term memory networks are used in fields as disparate as robot control (Mayer et al., 2006), protein homology detection (Hochreiter, Heusel and Obermayer, 2007), and rainfall detection (Shi et al., 2015). We use a modification of long short-term memory networks called a gated recurrent unit (Cho et al., 2014). RNNs and other deep learning architectures like convolutional neural networks have been used to forecast unemployment (Smalter Hall and Cook, 2017). Within economics, gated recurrent units (GRUs) have been applied in stock market prediction (Minh et al., 2018) and power grid load forecasting (Li, Zhuang and Zhang, 2020).

Moving from machine learning models to economics models, autoregressive models have been the workhorse forecasting models since the late 1930s (see Diebold (1998) and Walker (1931)). Even the machine learning models maintain an autoregressive structure in its inputs. Despite its simplicity and age, the model is still used among practitioners and as a baseline in many papers (Watson, 2001). One advancement in forecasting stems from the greater adoption of structural or pseudo-structural time series models like the Smets-Wouters DSGE models (Smets and Wouters, 2007). While DSGE forecasting is widely used in the literature, they are competitive with, but often times no better than, a simple AR(2), with more bespoke DSGE models performing poorer (Edge, Kiley and Laforte, 2010). However, the use of DSGE models for counterfactual analysis is an important and unique benefit of these models. The final economic baseline is the factor model (Stock and Watson, 2002*a*). The factor model attempts to use a large cross section of data which results in a more comprehensive picture of the economy to perform forecasting. Details on all these models and our implementation can be found in Appendix D. In addition, our paper

uses tools from forecast evaluation (West, 1996), (Pesaran and Timmermann, 1992), and (Diebold and Mariano, 2002), as well as model averaging (Koop, Leon-Gonzalez and Strachan, 2012), (Timmermann, 2006), (Wright, 2008).

Moving to structural economics, there is a small but robust literature data on panel data and dynamic general equilibrium models (Breitung, 2015). Most of the literature focuses on the use of panel data to better identify effects of interest that vary across country. Much of it is theoretical and adopts microeconometric panel techniques to macroeconomic data (Banerjee, Marcellino and Osbat, 2004). This literature also studies the use of cross sectional data to improve counterfactual analysis in general equilibrium models(Miyamoto, Nguyen and Sergeyev, 2018) (Crandall, Lehr and Litan, 2007) or have a more microeconomic forecasting focus (Baltagi, 2008). There is also a literature looking at specific panel models applied to macroeconomics like dynamic panel models (Doran and Schmidt, 2006), (Bai and Ng, 2010), (Diebold, Rudebusch and Aruoba, 2006). While much of the literature focuses on estimating different models, with fixed effects for instance, on different panel dimensions, there is influential work which theoretically supports to use of estimating a single set of parameters across the entire panel (Pesaran and Smith, 1995). In contrast to a focus on internal validity/identification, we focus on external validity, which is a benefit modelers can apply even if the phenomena of interest requires only the data of one country. However, our results hint at internal validity benefits as well. Likewise, since predictive power is a good proxy of external validity, we focus on empirically testing predictive power of the use of panel data.

III. Machine Learning Models

III.A. Automated Machine Learning

In order to provide a proxy for the performance of a good "nonparametric machin" learning model, we tested the open-source automated machine learning (AutoML) software H2O[1]. AutoML software is designed to provide end-to-end solutions for machine learning problems by efficiently training and evaluating a number of different models and ultimately returning the best model. We created an AutoML pipeline for each prediction horizon, trained the model on using our international cross-sectional data, evaluated on US data, and lastly predicted using our US data test set. Unlike with our own custom model, setting up H2O and training on our dataset was almost entirely automated. The benefit of automation is that while humans' can improve performance, there was little we could do, either via tinkering with architecture or devoting more computational resources to influence the performance of the procedure. The AutoML software picked a different model for each horizon (in order from predicting 1 quarter ahead to 5 quarters ahead): XGBoost, Gradient Boosting Machine, Gradient Boosting Machine, Distributed Random Forest, and Deep Learning. We noticed that the software generally picked deep learning models for the quarters that were further away (4 and 5 quarters ahead) compared to predicting gradient boosted techniques for closer quarters (1 and 2 quarters ahead). Ultimately, AutoML had very strong results and can be applied to other prediction problems in economics.

In addition, because the AutoML selects a different model for a given horizon and data set size, we also estimated a recurrent neural network on the both the reduced USA dataset and the expanded panel data set, which can illustrate

1. https://www.h2o.ai/

the effect of the increase in data size holding the model architecture fixed. The recurrent neural network also has the advantage of not being a model considered by AutoML.

III.B. Our Neural Network Model Architecture

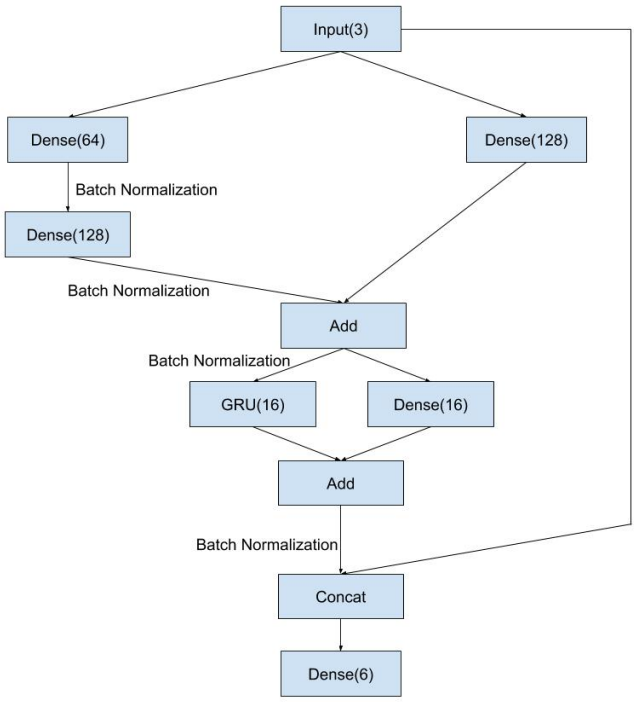

Figure I: Model Architecture

A recurrent neural network model we use to supplement AutoML is the gated recurrent unit (GRU) model, described in Appendix E.4. We add additional feedforward layers as well as other architecture choices as indicated in Figure I. The model architecture involves "pre-processing" the data using two feed-forward dense layers with 64 and 128 hidden units respectively and rectified linear unit (Agarap, 2018) activation (see Appendix E.5.), then running a gated

recurrent unit with 16 states on this "pre-processed" data[2]. Finally the output of the GRU is concatenated with the original input (lagged GDP, consumption and unemployment) and fed into a linear layer which forecasts an output.

One will notice that our model contains parallel dense layers between each operation. For example, there is a dense unit running in parallel with the two dense layers a feeding into the GRU(16), there is another Dense(16) that runs in parallel with the GRU, and before the final output is concatenated with the input and both fed into the final dense, skipping the entire network. These operations were originally skip connections (He et al., 2015), but were modified to allow for learning of linear parameters. The final skip connection concatenates the input with the output of the network so that the neural network would nest an VAR(1) model. These design choices all improved forecasting performance. Between all of our non-skip connection layers, we also use batch normalization (Ioffe and Szegedy, 2015). More details on batch normalization can also be found in Appendix E.6. Ultimately are model is about 17,000 parameters which explains the comparative outperformance on a data rich regime.

IV. Economic Models

We tested the external validity of a series of machine learning and traditional macroeconomic models estimate on our panel of countries. Additionally, we provide comparisons to the Survey of Professional Forecasters (None, 1968) median GDP forecast, which is seen as a proxy for state-of-the-art performance. A discussion of the Survey of Professional Forecasters and our attempt to evalu-

2. While we use the word pre-processed, the approach is trained entirely end-to-end and not a two step process as the word "pre-process" might imply. However this neural network, projects the input data (consumption growth, GDP growth and the unemployment rate) into a high dimensional state that the gated recurrent unit finds easier to work with much like pre-processing would. The end-to-end learning procedure learns the pre-processing and the gated recurrent unit analysis at the same time

ate their forecasts is contained in Appendix C. The baseline economic models we used are the AR(2) autoregressive model, the Smets-Wouters 2007 DSGE model (Smets and Wouters, 2007), and a factor model (Stock and Watson, 2002*a*) and (Stock and Watson, 2002*b*) and a VAR(1) (Sims, 1980). A more detailed explanation of these models is contained in Appendix D. For the reduced form models, getting cross country data is straightforward, thus we compare those models estimated only on US data as well as on our data set of 50 countries. For the Smets-Wouters DSGE, we also assembled a panel of 27 developed and rich developing countries to estimate the structural model on.

As is standard with economic forecasting, the baseline models are trained in a pseudo-out-of-sample fashion where the training set/validation set expands as the forecast date becomes more recent. However, with our neural network and AutoML, we keep the training set and validation set fixed due to computational costs and programming constraints. We suspect our model will likely improve if we used a pseudo-out-of-sample approach.

V. Data and Method

United States macroeconomic data is not sufficient enough to train a complex model like a neural network. To fix this, we use data from 49 other developed and developing countries as listed in Appendix A. We source cross country data from Trading Economics via the Quandl API[3] as well as GDP data from the World Bank[4]. We use GDP, consumption, and the unemployment rate as inputs to the model. GDP and consumption are all expressed in growth rates. Unemployment is expressed as a rate as well. As mentioned earlier, we also assemble 11 different covariates across 27 countries for a panel of data used

3. https://www.quandl.com/tools/api
4. World Development Indicators, The World Bank

to estimate the Smets-Wouters DSGE. Data comes from FRED, World Bank, Eurostat, OECD and the IMF.

We split our data into training, validation, and test sets. We forecast GDP, evaluated with root mean squared error (RMSE). The test set is from 2008-Q4 to 2020-Q1, either testing only US data or world data depending on the particular problem. The validation consists of data from 2003-Q4 to 2008-Q3, which is only used for the recurrent neural network. This data is in the training set for all other models. AutoML, which is not a sequential model, uses k-fold cross validation on the entire training. The training set is all other periods and depends on when each country started releasing their data. We chose these periods so that both the test set and the validation set would have both periods of expansion and recession, based on the US business cycle. Including the 2001 recession in the validation set would leave the model without enough training data, so we split the data of the Great Recession over the test and validation set. The quarter with the fall of Lehman Brothers and the largest dip in GDP was the first quarter in our test set, 2008-Q4. There are two quarters with negative growth preceding this which are in the validation set. We estimated all models from a horizon of 1 quarter ahead to 5 quarters ahead. The metric of choice for forecast evaluation is RMSE.

VI. Results

Our first set of results deal with applying data augmentation to reduced form models like VARs and AR models and showing improved GDP forecasting accuracy. Our second set of results show that the panel data augmentation improves external and potentially internal validity of structural models, using the Smets-Wouters DSGE model as the main benchmark. Our third set of results take all the models and demonstrates the forecasting power of "nonparametric"

machine learning models over all the above traditional economic models in this relatively data rich regime.

VI.A. Reduced Form Models

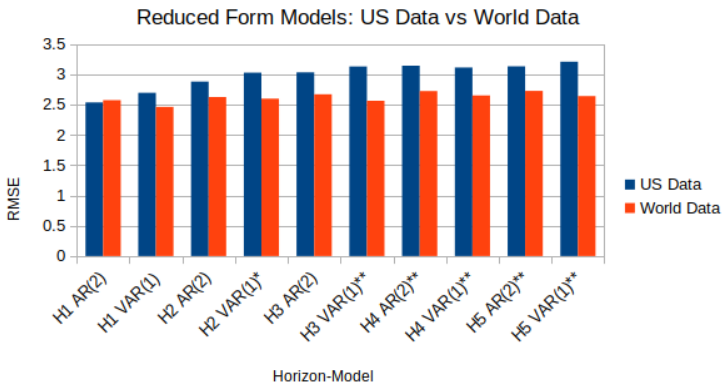

FIGURE II: EVALUATION OF RMSE FOR LINEAR MODELS USING BOTH US ONLY DATA AND CROSS-COUNTRY PANEL DATA

Figure II shows the forecasting performance of either a VAR(1) or AR(2) model at a particular forecast horizon. The first bar in each pair represents the forecast RMSE on the test set (2008Q4-2020Q1) with the model estimated only on US data. The second bar in each pair represents forecasting performance using the model estimated on the entire panel of countries. The stars next to the model name indicates that world data outperformance is significant using a Diebold-Mariano test (Diebold and Mariano, 2002) at the 1% (***), 5% (**) and 10% (*) level. This format will be followed for the most of the RMSE performance graphs. Especially at the longer forecast horizons, the world panel data improves the performance of the models in statistically significant manner. Except for a slight underperformance at the 1 quarter ahead for the AR(2), all other horizon-models showed outperformance using the country panel data augmentation. The outperformance of the world panel averaged roughly 12%

of US RMSE over all horizon-model pairs.

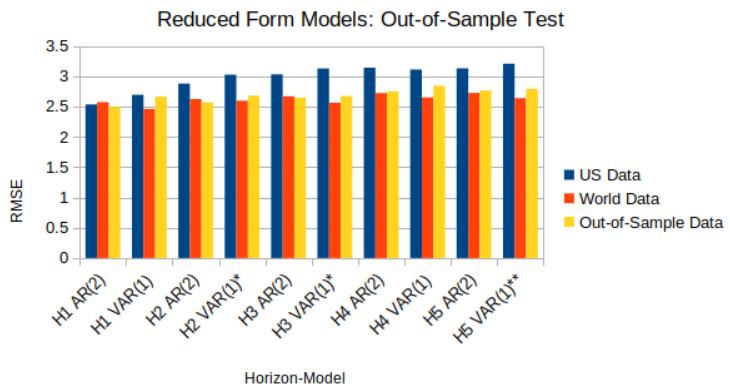

Figure III: Evaluation of Out-of-Sample RMSE for linear models

Figure III shows similar data to Figure II, except the third bar for each horizon-model triplet shows forecasting performance on the same US test data, but for a model that is both time step and country out-of-sample. For example, since we are forecasting US GDP, we use every country but the US data to forecast US GDP. This test enables us to show that using panel data can lead to models that are policy/country invariant and can generalize even to country data that the model has not had data access to. In all cases, the RMSE of the out-of-sample forecasts are statistically indistinguishable from the RMSE of the model estimated on the full panel of countries. Stars indicate significance of the out-of-sample forecast over just using the United States training data. Again the models are mainly significant at longer horizons, but any significance is impressive since the outperforming model uses no US data to forecast US GDP. Excluding the H1 AR(2) pair, using only the out-of-sample data leads to capturing, on average, 79% of the outperformance of the world panel over using only US data. We also provide VAR(4) performance data using the same approach along with all the above data in tabular form in the Appendix G.1.

TABLE I: Average Forecasting Performance Evaluated over the World

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
AR(2)					
Local Data	5.22	5.33	5.51	5.56	5.60
World Data	4.88	4.98	5.10	5.19	5.19
Out-of-Sample Data	5.07	5.13	5.36	5.36	5.38
VAR(1)					
Local Data	5.10	5.17	5.19	5.29	5.26
World Data	4.80	4.94	5.04	5.11	5.12
Out-of-Sample Data	4.92	5.00	5.07	5.20	5.25
VAR(4)					
Local Data	7.90	7.05	7.74	7.87	9.27
World Data	4.72	4.90	5.03	5.10	5.11
Out-of-Sample Data	4.70	4.87	5.02	5.17	5.26

* Hypothesis testing has not been performed on this data yet

Finally in Table I, as an additional robustness check, we perform the same tests over our entire cross section of countries. For example, for each local forecast, we only use French data to forecast French GDP. Likewise for world data, we use the entire data set to estimate a single model, then for each country we use the same model to jointly forecast the GDP of every country and average RMSE across countries. For out-of-sample data, a different model is trained to forecast each country, but each model is estimated using every country but the country of forecasting interest. Aside from providing an additional robustness test regarding the outperformance of world data and even out-of-sample data, this table shows additional policy invariance both from the out-of-sample tests as well as the world data tests, where the same model can jointly forecast all countries much better than custom tailored models for each country. Additionally, this model seems to indicate a diminishing return in terms of RMSE for linear models as the data gets more plentiful and it seems like all linear models converge to a similar RMSE.

VI.B. Structural Models

Considering the success of the data augmentation for reduced form model, we also test the procedure on structural models and achieve similar success. Since the DSGE model requires 11 different variables, we assembled our own data from the World Bank, OCED, Eurostat and FRED. This panel has only 27 countries cross section which is smaller than the 50 country panel used in our reduced form models. Our first result just compares the forecasting performance DSGE model trained only on US data with a DSGE model estimated on the entire panel of countries in the same manner as Figure II. Both models underperform the parameters recovered in the original Smets and Wouters (2007), because we focus on maximum likelihood estimation without priors optimized only with gradient descent. Despite the forecasting success of (Smets and Wouters, 2007) and others Bayesian DSGE models (Herbst and Schorfheide, 2015), (Fernández-Villaverde, Rubio-Ramírez and Schorfheide, 2016), we choose to use to use a maximum likelihood approach to maintain comparability to the reduced form and machine learning experiments as well as a large portion of the applied literature which focus on point estimate techniques ranging from calibration, maximum likelihood to generalized method of moments (Hansen and Singleton, 1982). Not to undersell our methodology though, Figure V as well as Appendix G.2. shows the performance of a model estimated via maximum likelihood that outperforms the parameters of Smets and Wouters (2007) along some of the horizons and likely more could be done if this was the central objective of our paper.

Unlike the reduced form models, with the DSGE, the most significant results with p-values less than 1% are at shorter horizons below 3 quarters ahead, although all horizons are significant at the 10% level. Surprisingly, given the Lucas critique and the inductive biases of these models, the improvement in

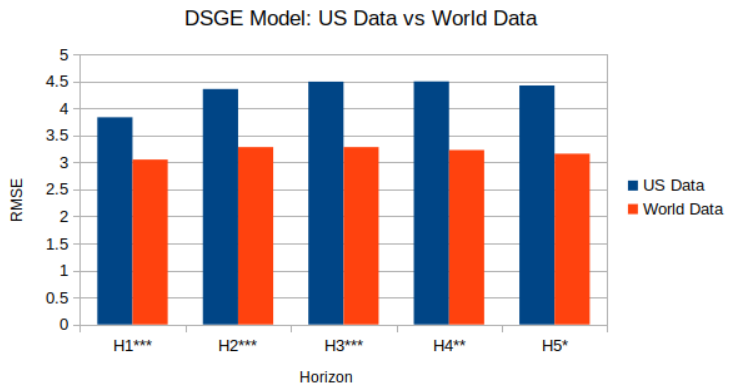

Figure IV: Evaluation of RMSE for DSGE models using both US only data and cross-country panel data

RMSE averages over 25% over all horizons, which is more than double the percentage improvement for reduced form models. Part of the outperformance is due to significantly weaker performance estimated only on US data. However, this suggests that the Smets-Wouters model is at least no better at generalizing across policies or countries.

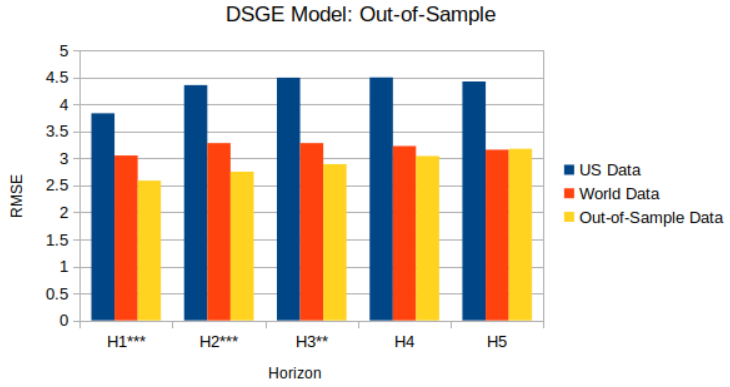

Figure V: Evaluation of Out-of-Sample RMSE for DSGE models

In Figure V, we also provide a structural chart the parallels the out-of-sample chart in Figure III in the reduced form section. As a reminder, the out-of-sample

DSGE model is estimated on the panel of 26 countries–removing the US–so that the GDP forecast is country out-of-sample as well as time step out-of-sample. The out-of-sample performance is on average 9% better than even the performance of a DSGE estimated on the entire world data. Despite that, the Diebold-Mariano tests are less significant with only the first two horizons having p-values with less than 1% and 5% for 3 horizons and no significance in horizon 4 and 5. This suggests potentially that the out-of-sample outperformance may in part be due to luck, but this improvement we study this in further detail. Since the United States is the only country that has data that extends much before 2000, we hypothesize that DSGE model parameters are more stable across countries than across time and when one removes the US, one makes the data much more similar to the test set across time. This addresses an internal validity criticism of our panel approach which argues that our structural parameters have different means across countries and when estimating a single model on all countries the recovered values lose much economic meaning (Pesaran and Smith, 1995). After all, the only large country in the world is the United States (and maybe China and the EU if you expand the definition of country), and so estimating the large country Smets-Wouters model on Spanish or French data will yield parameters that by assumption don't correspond to actual economic primitives. The improvement of performance after taking the US from the training set could suggest that structural parameters are more similar across space than across time. Thus using our panel data can result in models with both more internal validity and external validity . Less historical data can be used if a larger panel of countries is used, which could be the more favorable trade off for structural models.

Probing this hypothesis leads to results that are somewhat more mixed. We estimate a model trained on the entire panel of countries with data from 1995Q1

onward. This only effected three countries–the US, Japan and New Zealand. This procedure more sharply isolates the effect of similarity across space and across time on model generalization, than just removing all US data. The US lost about 140-190 data points (as the test set requires rolling forecasts), and New Zealand and Japan both lost about 15-60 time steps. Figure VI, illustrates this experiment to compare the performance of models estimated on data since 1995 to models estimated with full country and time steps.

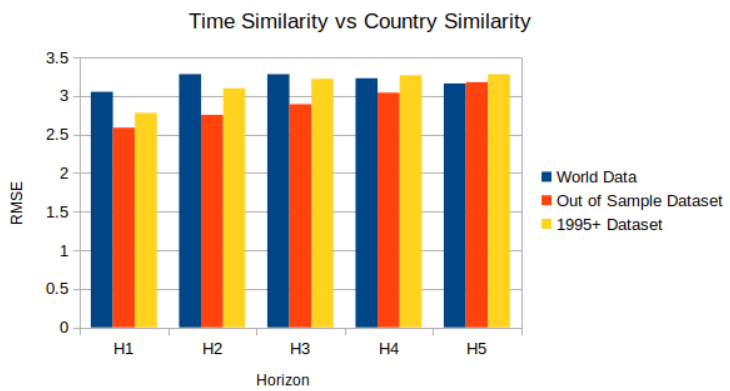

Figure VI: Evaluation of RMSE for linear models using both US only data and cross-country panel data

Figure VI shows GDP forecasts on the same test set 2008Q4-2020Q1. The first bar in each triplet is the DSGE model performance estimated using world data across all periods in time. The second bar is the out-of-sample performance as shown in Figure V. The third bar is the performance of a DSGE model estimated on all country data, but only since 1995 to make the data more relevant across time to the test set. It looks like the 1995 data performs worse than the out-of-sample test which could suggest some of the outperformance of the out-of-sample DSGE model as due to luck. However, it does look like the 1995 data performs at least just as well as a model estimated on world data both pre and post 1995. The results seem inconclusive but perhaps suggest

that estimated DSGEs models with data from different countries but certainly doesn't suggest any more parameter stability across time than space, in contrast with the potential pitfall highlighted by Pesaran and Smith (1995).

In closing, we show that using our panel data augmentation results in the Smets-Wouters DSGE model outperforming DSGE models estimated only on US data. Furthermore, we provide suggestive evidences that it may be possible that internal validity is better if one uses data across countries in addition to the statistical significant improvement in external validity. Using our out-of-sample tests, we show that these models can improve the out-of-sample generalization of the Smets-Wouters DSGE model even if they are theoretically policy invariant. The data shown in the charts will also be displayed in a table in Appendix G.2. As a final note, while it is difficult to quantify improved performance of calibration and generalized methods of moments, based on the external validity improvements from estimation, we imagine macroeconomists would benefit from calibrating to moments that feature a large number of countries.

VI.C. Nonparametric Machine Learning Models

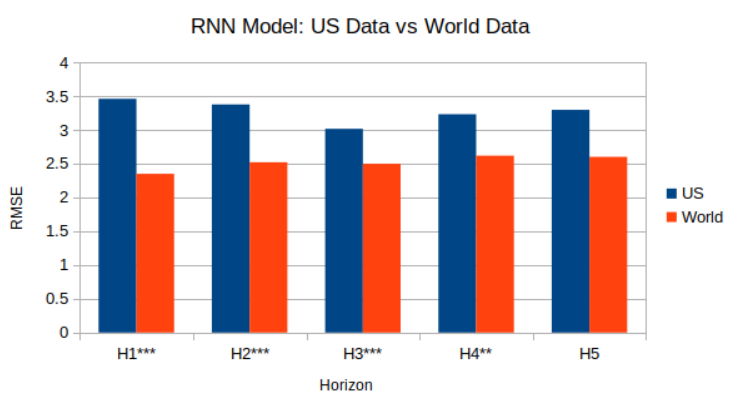

Figure VII: Evaluation of RMSE for RNN models using both US only data and cross-country panel data

We repeat the same data augmentation test for both the recurrent neural network and models produced by AutoML. For the recurrent neural network, Figure VII shows the improvement in RMSE from estimating a recurrent neural network using only US data to using the entire cross section of 50 countries. The improvement is statistically significant for all horizons except five quarters ahead. The average improvement was around 23% over all horizons, which is similar to the improvement for the Smets-Wouters model and almost double the improvement of linear models. The improved performance of the Smets-Wouters model when moving to the country panel is partially due to a lower US performance baseline. However, the recurrent neural network performs worse on US data than comparable linear models then performs better when using world data, but better than the DSGE model estimated only on US data. This is a reassuring confirmation as the recurrent neural networks is a data hungry "nonparametric" model which benefits more from data rich regimes than linear models.

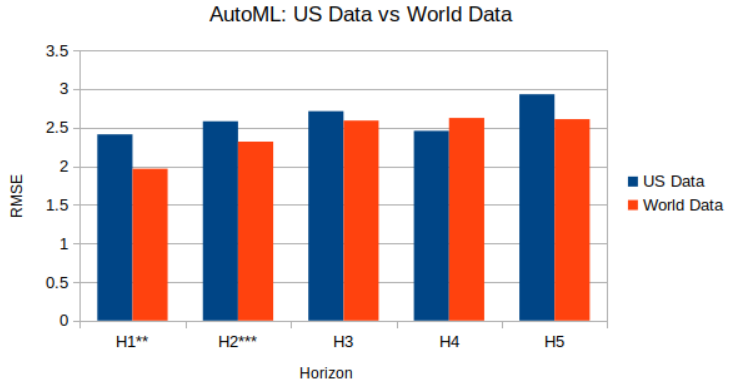

Figure VIII: Evaluation of RMSE for AutoML using both US only data and cross-country panel data

Our second chart in Figure VIII, shows the same performance graph for AutoML. The performance gain is not as easily interpreted as just coming from

additional data as the increase in data also resulted in more data-hungry and flexible models outperforming in the panel data regime. AutoML has the least amount of improvement in performance when using the panel of countries as training data, with average improvements in RMSE of about 7.5 percent. Only the first two horizons show statistically significant forecast performance. Much of this is likely due to the shift of models to low variance/high bias models in the data poor regime to high variance/low bias models in the data rich regime. Unsurprisingly, it is the sample efficiency of low variance models that reduces the gap small between the two data regimes. However, it's worth pointing out that the improvements are still significant along some horizons, especially in light of nearly state of the art performance using only US data to begin with. Even so, only in four quarters ahead does the AutoML model outperform all the baseline models trained on US data. When estimated on world data, AutoML outperforms all economic models on all horizons except three quarters ahead and it's performance at one quarter ahead rivals the Survey of Profession Forecasters despite being estimated on lags of GDP, consumption, and unemployment.

Table **??** below compares the forecasting performance of all models, both estimated on a panel or just the US. This table demonstrates the effectiveness of the machine learning forecasting methods in data rich regimes. AutoML estimated on world data outperforms all baseline economic models on four of the five horizons and the RNN outperforms on longer term horizons. We do not report the maximum likelihood Smets-Wouters model, as the original Bayesian parameterization has better performance than either of our maximum likelihood Smets-Wouters models estimated on world or US data. We could have included our out-of-sample Smets-Wouters model, which would improve DSGE performance but would not change anything substantive in the horse race. Regardless, we bold any economic models that outperform all our machine learning models

and bold and machine learning models that outperform all economic models. It seems the best performing model along all horizons was either an AutoML model or an RNN model.

TABLE II: RMSE of Our RNN, AutoML, and Baseline Models

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
VAR(4)					
US Data	2.99	3.03	3.10	3.08	3.08
World Data	2.37	2.52	2.56	2.63	2.63
AR(2)					
US Data	2.53	2.88	3.03	3.14	3.13
World Data	2.57	2.62	2.67	2.72	2.72
Smets-Wouters DSGE Bayesian					
US Data	2.79	2.95	2.89	2.80	2.71
Factor					
US Data	2.24	2.48	2.50	2.67	2.86
RNN (Ours)					
US Data	3.46	3.37	3.01	3.23	3.30
World Data	2.35	2.52	**2.50**	**2.62**	**2.60**
AutoML (Ours)					
US Data	2.41	2.58	2.71	**2.45**	2.92
World Data	**1.97**	**2.32**	2.59	**2.62**	**2.61**
SPF Median	1.86	2.11	2.36	2.46	2.65

We also provide a graph of the forecasts of five of the models: AR(2), factor, DSGE, RNN, and AutoML, as well as the true data. We display the one quarter ahead forecasts in Figure IX here and the rest of the graphs are in Appendix B.

AutoML, factor, and RNN models all do a good job at forecasting the Great Recession, with AutoML forecasting the best at 1 quarter ahead. The AR(2) and DSGE do not detect a regime change for the recessionary periods and are also upwardly bias leading to even worse performance during recession onset[5]. However, the XGBoost model that performs the best in AutoML is able to do a good job at forecasting the expansions both in terms of the average level as well is individual movements in the quarterly data. Neither the factor model nor the RNN are able to forecast the quarter by quarter movements with such

5. More information on the biases and variances of the different models can be found in Appendix F.2.

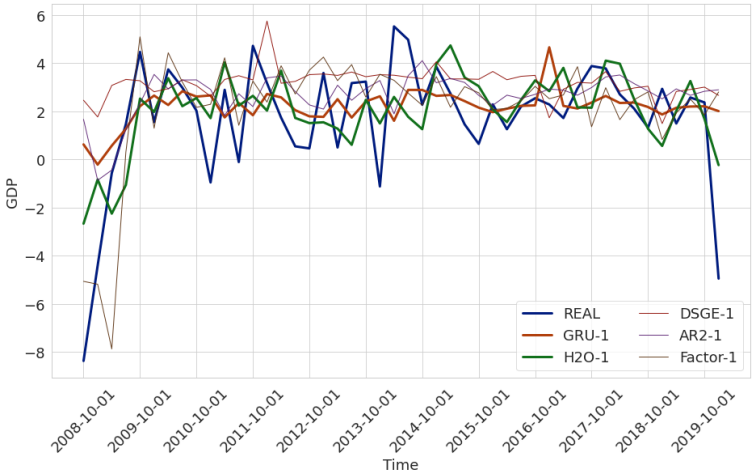

Figure IX: 1-Quarter ahead - Forecasts

accuracy. More information about the performance especially of our RNN model are labeled in our robustness checks in Appendix G.3.

VII. Conclusion

In this paper we showed how estimating macroeconomic models on a panel of countries, as opposed to a single country can significantly improve external validity. Using a panel of countries as a training set, we statistically significantly improved the RMSE performance of reduced form models–AR(2), VAR(1), and VAR(4)–by roughly 12 percent. We further showed that we can make these reduced form models more policy/country invariant, suggesting that these models have learned to generalize GDP forecasting even to countries the model has never been trained on.

Next we show that a similar training set of a panel of countries can improve external validity of structural models which again are typically estimated only on a single country of interest. We focus on the Smets-Wouters model (Smets and Wouters, 2007). Using a panel of countries improves the forecasting

performance of the Smets-Wouters model estimated with maximum likelihood by roughly 24% averaged across horizons. These results are again statistically significant. We then demonstrate that we can again improve policy invariance and generalization to out-of-sample countries by using a panel of countries in our training set. In contradiction with the Lucas critique, the out-of-sample improvement for the Smets-Wouters model is larger than reduced form models. One potential roadblock to adopting a panel of country estimation approach is the fact that the structural parameters may not be stable across countries. While our results are less conclusive on this front, we argue based on a forecasting exercises, that parameter generalization and stability are likely as good across space as across time. We imagine our results will extend to other estimation techniques like generalized method of moments, calibration, and Bayesian approaches.

Finally, recognizing that our dataset has increased from 300 time steps to around 3000 time step-countries, we show that "nonparametric" machine learning models are able to outperform all the economic baseline models even after being estimated in this more data rich regime. Our recurrent neural network outperforms all economic baselines for horizons longer than two periods ahead. Likewise, our AutoML model outperforms all baselines for all horizons except for the three quarters ahead. Combined, the best performing model over all horizons is either an AutoML model or a recurrent neural network model which suggests there is likely much room to test other non-parametric models in the more data rich macroeconomic regime. For future work to look into using these models to interrogate economic phenomenon.

VIII. Appendix

A Selected Countries

Countries in reduced form data set: Australia, Austria, Belgium, Brazil, Canada, Switzerland, Chile, Columbia, Cyprus, Czech Republic, Germany, Denmark, Spain, Estonia, European Union, Finland, France, Great Britain, Greece, Hong Kong, Croatia, Hungry, Ireland, Israel, Italy, Japan, Korea, Luxembourg, Latvia, Mexico, Mauritius, Malaysia, Netherlands, Norway, New Zealand, Peru, Philippines, Poland, Portugal, Romania, Russia, Singapore, Slovakia, Slovenia, Sweden, Thailand, Turkey, USA and South Africa.

Countries in structural data set: Australia, Austria, Belgium, Canada, Chile, Columbia, Germany, Denmark, Spain, Estonia, Finland, France, Iceland, Israel, Italy, Japan, Korea, Lithuania, Luxembourg, Mexico, Netherlands, New Zealand, Poland, Portugal, Slovakia, Slovenia, Sweden, USA

B Selected Performance: Graphs

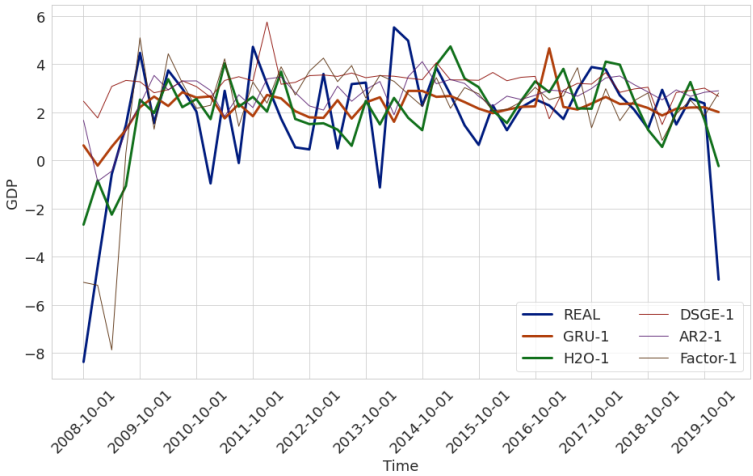

Figure X: 1-Quarter ahead - Forecasts

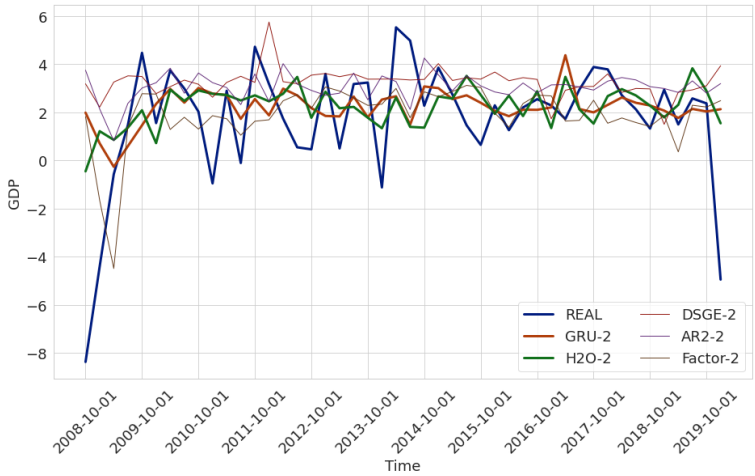

Figure XI: 2-Quarters ahead - Forecasts

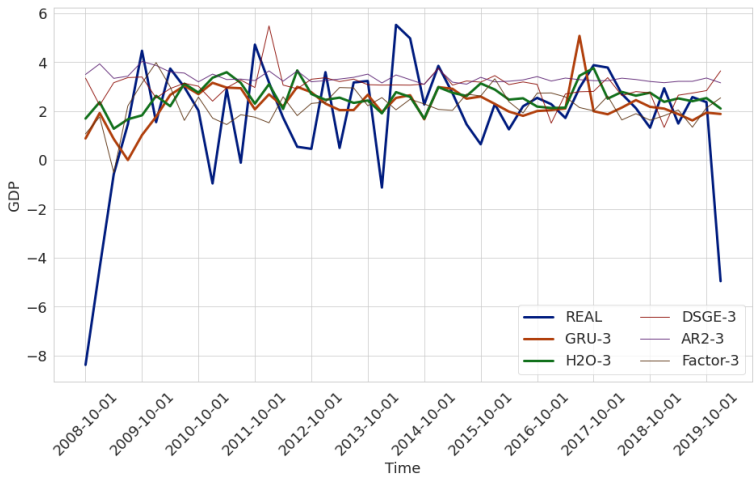

Figure XII: 3-Quarters ahead - Forecasts

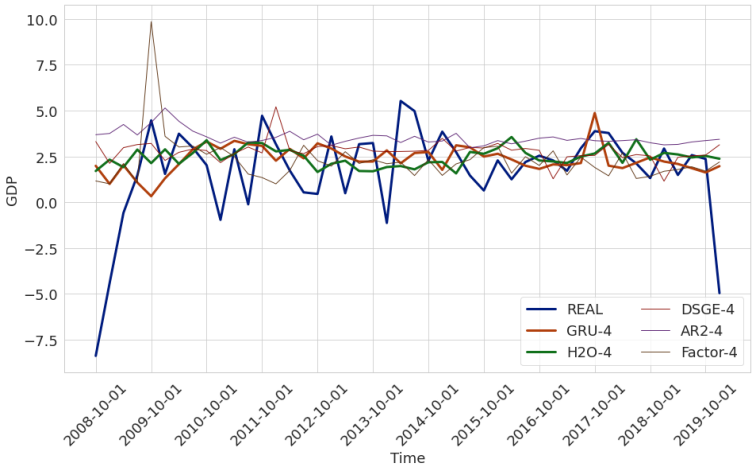

Figure XIII: 4-Quarters ahead - Forecasts

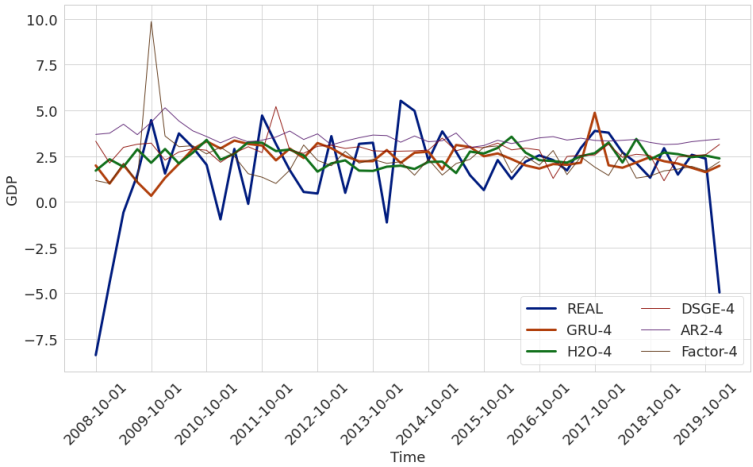

Figure XIV: 5-Quarters ahead - Forecasts

C Details on the Survey of Professional Forecasters

While our model uses the 2020 vintage of data, in reality, the forecasters for the Survey of Professional Forecasters were working with pseudo-out-of-sample vintages when forecasting over the entire test. While reproducing this would be possible from using old vintages, this would require estimating the model at every time step of the test set as the data would change every period. We wanted to avoid this pseudo-out-of-sample forecasting as it would result in estimating 4600 models instead of 100 at each horizon. Furthermore, the benefit of this increased computation is not clear as we would still be using 2020 vintage data for countries outside the US, many of which old vintages are difficult or impossible to find. As such, it was easier to compare the SPF performance on the 2020 vintage. Furthermore, all our baselines models are estimated and evaluated using the 2020 vintage as well and so this choice allows one to compare the SPF performance with the performance of all the baseline models.

D Detailed Description of Economic Baseline Models

A first model we use is the autoregressive model, AR(n). An often used benchmark model, it estimates a linear relationship using the independent variable lagged n times. In terms of forecasting ability, this model is competitive with or outperforms the other economic models in our tests which is consistent with Diebold (1998). We use an autoregressive model with two lags and a constant term.

Additionally, we compared the Smets-Wouters 2007 model (Smets and Wouters, 2007), as DSGE models share many similarities with recurrent neural networks and the Smets-Wouters paper cited directly above suggests that this particular model can outperform VARs and BVARs in forecasting. When running this, we use the standard Smets-Wouters Dynare code contained on the published pa-

per's data appendix. For computational reasons, we estimate the model using Laplace's approximation to the optimal Bayesian posterior and then forecast based on the posterior of the deep parameters.

A final model we included in our baseline economic models were factor models (see Stock and Watson (2002*a*) and Stock and Watson (2002*b*)). While these models were the most effective especially at lower horizons, these models are also dependent on a large cross section of economic data with a long history in a country. In reality, only a few other developed countries have a cross section of data that would be large enough to permit using these models as effectively as can be used in the United States. That being said, factor models do outperform our neural networks at shorter time intervals, and we imagine there is promise in combining the factor approach with a neural network approach. In short, the factor model approach takes a large cross section of data and uses a technique like principal components analysis to reduce the dimensionality of the problem. In our case, we concatenate 5-8 principle components based on information criterion of the high dimensional data with a lagged value of GDP and regress future GDP on this. We modify and use the code from FRED-QD as our baseline factor model (McCracken and Ng, 2016).

While we considered the forecasting performance of vector autoregressions (Sims, 1980), these models did not perform well enough to report in our main tables. However, because of available cross country data, we compared this model and the AR(2) with our GRU on our 50 countries cross section test. Since we are only forecasting GDP, the vector autoregressive models use lagged GDP, consumption, and unemployment used to forecast the single GDP variable as a linear regression.

All the economic models were estimated on US GDP as is standard. While we ran preliminary tests on estimating these models on our cross section of 50

countries, we ran into issues with estimating both factor models and DSGE models this way. However, preliminary results on the AR(2) model suggests there could be some improvement to using a cross section even on a 3 parameter AR(2) model. The improvement is not as large as the GRU, which is not surprising as the GRU has more parameters to take advantage of a larger data set.

E Neural Network Models

E.1 Feed-Forward Neural Networks

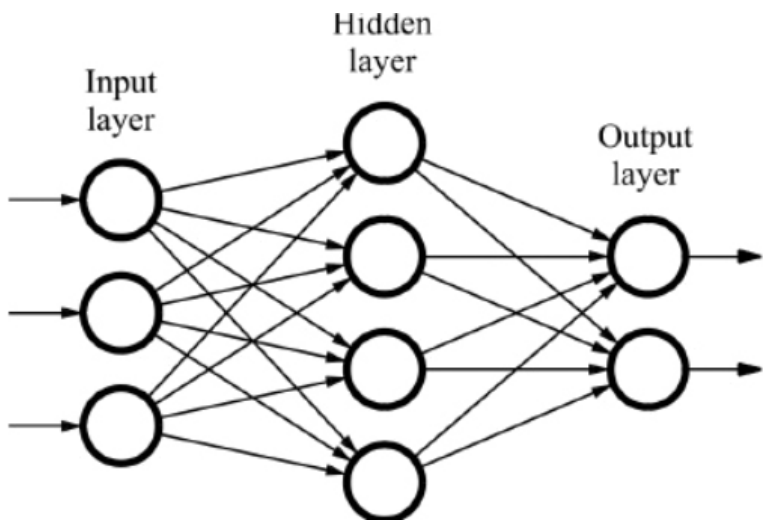

Figure XV: An Example of a Feed-Forward Neural Network

The feed-forward network is the prototypical image associated with deep learning. At its core, a feed-forward neural network is a recursively nested linear regression with non-linear transforms. For example, assume X is a vector valued input to the neural network and X_{out} is the output. In a typical linear regression, $X_{out} = X_{in}\beta_1$. The insight for composing a feed-forward network is to take the output and feed that into another linear regression: $Y = X_{out}\beta_2$.

In Figure XV, X_{in} would be the input layer, X_{out} would be the hidden layer and Y would be the output layer. The problem is not all that interesting if X_{out} is a scalar. If X_{in} is a matrix of dimension: time steps by regressors, X_{out} can be a matrix of dimension time steps by hidden units. Here in the figure, the dimension of the hidden layer is 4. Thus, β_1 has to be a matrix of dimension 3 by 4 (regressors by hidden units). Thus, we make X_{out} an input into multidimensional regression for the second layer, $Y = X_{out}\beta_2$, if the first layer is a vector regression[6]. This can be repeated for as many layers as desired.

Now a composition will result in: $Y = X_{out}\beta_2 = (X_{in}\beta_1)\beta_2$. A product of two matrices is still another matrix, which makes this model still linear regardless of how many layers there are. However, an early result in the literature showed that if between every regression, eg $X_{out} = X_{in}\beta_1$, one inserts an almost arbitrary non-linear link function this allows a neural network to approximate any continuous function (Hornik, Stinchcombe and White, 1989). For example, inserting a logistic transformation between X_{in} and X_{out} i.e. $X_{out} = \sigma(X_{in}\beta_1)$ where $\sigma(z) = \frac{1}{1+e^{-z}}$. One can put these non-linearities as often as one would like to get something like this: $Y = \sigma(\sigma(X_{in}\beta_{1,1})\beta_2)$. These are the fundamental building blocks of neural networks and additional composition of link functions and matrix multiplication of parameters form the basis of deeper networks and allows these models to be universal approximators.

E.2 The Simplest Recurrent Neural Network: A Linear State Space Model

For the purposes of this paper, conceptualizing a recurrent neural network as a distinct model from feed-forward networks will not hinder comprehension. That being said, without even knowing it, many economists are already familiar

6. Note: this regression is not a vector autoregression as X_{out} is a latent variable

with recurrent neural networks. The simplest is a Kalman filter-like linear state space model. The two equations that define the linear state space model are[7]:

$$\mathbf{s}_t = \mathbf{s}_{t-1}\mathbf{U}^s + \mathbf{y}_{t-1}\mathbf{W}^s + \mathbf{b}^s, \tag{1}$$

$$\mathbf{y}_t = \mathbf{s}_t\mathbf{U}^y + \mathbf{y}_{t-1}\mathbf{W}^y + \mathbf{b}^y \tag{2}$$

In a linear state space model, the state $\mathbf{s}_i$ is an unobserved variable which allows the model to keep track of the current environment. One uses the state, along with lagged values of the observed variables, to forecast observed variables $\mathbf{y}_i$. For example, for GDP, the state can be either a expansionary period or a recession. A priori, the econometrician does not know. However, one can make an educated guess based on GDP growth. As machine learning is more interested in prediction, the state is estimated with point estimates, which allows the data scientist to sidestep the tricky problem of filtering.

The problem with linear state space models is that if one does not apply filtering, the state vector either blows up or goes to a steady state value. This can be seen by recognizing that each additional time step results in the state vector getting multiplied by $\mathbf{U}^s$ an additional time. Depending on if the eigenvectors of $\mathbf{U}^s$ are greater than or less than one, the states will ultimately explode (go to infinity) or go to a steady state. More sophisticated neural networks like gated recurrent units (Cho et al., 2014) we use, fix this with the use of gates.

7. We add autoregressive lags to make the model more general.

E.3 Estimating the Parameters of a Linear State Space Model on Data

Going back to the two equations that define the linear state space model:

$$Y_t = D * S_t + E * Y_{t-1} + F, \tag{3}$$

$$S_t = A * S_{t-1} + B * Y_{t-1} + C \tag{4}$$

We use Equations (3) and (4) to recursively substitute for the model prediction at a particular time period. The forecast for period 1 then is:

$$\hat{y}_1 = D * (A * 0 + B * Y_0 + C) + E * Y_0 + F \tag{5}$$

and the forecast for period 2 is:

$$\hat{y}_2 = D * (A * (A * 0 + B * Y_0 + C) + B * Y_1 + C) + E * Y_1 + F \tag{6}$$

Hatted variables indicate predictions and un-hatted variables correspond to actual data. Additional time periods would be solved by iteratively substituting for the state using Equations (3) and (4) for the previous state. In order to update the parameters matrices A, B, C, D, E, and F, the gradient is derived for each matrix and each parameter is updated via hill climbing. We will illustrate the process of hill climbing by taking the gradient of one parameter B:

$$\frac{\partial \sum_{\forall t} L(y - \hat{y})}{\partial B} = \frac{\partial L(y_1 - \hat{y}_1)}{\partial B} + \frac{\partial L(y_2 - \hat{y}_2)}{\partial B} \tag{7}$$

Here $L()$ indicates the loss function. Substituting for y_1' and y_2' with Equations (5) and (6) into (7) and using squared error as the loss function, we arrive at an equation with which we can take partial derivatives for with respect to A:

$$\begin{aligned}(8)\quad \frac{\partial}{\partial B}L &= \frac{\partial}{\partial B}\frac{1}{2}(y_1 - D*(A*0 + B*Y_0 + C) + E*Y_0 + F)^2 \\ &+ \frac{\partial}{\partial B}\frac{1}{2}(y_2 - D*(A*(A*0 + B*Y_0 + C) + B*Y_1 + C) + E*Y_1 + F)^2\end{aligned}$$

Distributing all the B's and taking the derivative of (8) results in $\frac{\partial}{\partial B}L = -(y_1 - D*(A*0 + B*Y_0 + C) + E*Y_0 + F)*D*Y_0 - (y_2 - D*(A*(A*0 + B*Y_0 + C) + B*Y_1 + C) + E*Y_1 + F)*(D*A*Y_0 + D*Y_1)$ which provide the gradients for hill climbing. In practice, the derivatives are taken automatically in code.

E.4 Gated Recurrent Units

Gated recurrent units (Cho et al., 2014) were introduced to improve upon the performance over recurrent neural networks that resembled linear state space models and can deal with this exploding gradient problem discussed above. First we redefine sigma as the logistic link function:

$$(9)\qquad \sigma(x) = \frac{e^{\beta x}}{1 + e^{\beta x}}$$

The idea behind the gate, is to allow the model to control the magnitude of the state vector. A simple gated recurrent neural network looks like the linear state space model with an added gate equation:

$$\mathbf{y}_t = \mathbf{h}_t \mathbf{U}^y + E * \mathbf{y}_{t-1} \mathbf{W}^y + \mathbf{b}^y \tag{10}$$

$$\mathbf{z}_t = \sigma(\mathbf{h}_{t-1} \mathbf{U}^h + \mathbf{y}_{t-1} \mathbf{W}^h + \mathbf{b}^h) \tag{11}$$

$$\mathbf{s}_t = \mathbf{h}_{t-1} \mathbf{U}^s + \mathbf{y}_{t-1} \mathbf{W}^s + \mathbf{b}^s \tag{12}$$

$$\mathbf{h}_t = \mathbf{z}_t \odot \mathbf{s}_t \tag{13}$$

The output of $\sigma()$ is a number between zero and one which is element-wise multiplied by $\mathbf{s}_t$ which is the first draft of the state. The operation $\odot$ indicates element-wise multiplication or the Hadamard product. Variables are subscripted with the time period they are observed in (t or $t-1$). Weight matrices, which are not a function of the inputs, are super-scripted with the equation name they feed into. All elements are considered vectors and matrices, and matrix multiplication is implied when no operation is present.

The presence of the gate controls the behavior of the state, which means that even if the eigenvalues of $\mathbf{U}^s$ were greater than one, or equivalently even if $\mathbf{h}_t$ would explode without the gate, the gate can keep the state bounded. Additionally, the steady state distribution of the state does not have to converge to a number. The behavior could be periodic, or even chaotic (Zerroug, Terrissa and Faure, 2013). This allows for the modeling of more complex behavior as well as the ability of the state vector to "remember" behavior over longer time periods (Chung et al., 2014).

The equations of the gated recurrent unit are:

$$\mathbf{y}_t = \mathbf{h}_t \mathbf{U}^y + E * \mathbf{y}_{t-1} \mathbf{W}^y + \mathbf{b}^y \tag{14}$$

$$\mathbf{z}_t = \sigma(\mathbf{x}_t \mathbf{U}^z + \mathbf{h}_{t-1} \mathbf{W}^z) \tag{15}$$

$$\mathbf{r}_t = \sigma(\mathbf{x}_t \mathbf{U}^r + \mathbf{h}_{t-1} \mathbf{W}^r) \tag{16}$$

$$\mathbf{s}_t = \tanh(\mathbf{x}_t \mathbf{U}^s + (\mathbf{h}_{t-1} \odot \mathbf{r}_t) \mathbf{W}^s) \tag{17}$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{s}_t + \mathbf{z}_t \odot \mathbf{h}_{t-1} \tag{18}$$

Where tanh is defined as the hyperbolic tangent:

$$\tanh(x) = \frac{e^{2*x} - 1}{e^{2*x} + 1} \tag{19}$$

Like the linear state space model, the state vector of the gated recurrent unit persists over time steps in the model. Mapping these equations to Equation (10)-(13), Equation (18) is the measurement equation (analogous to Equation (10)). Equation (15) and (16) are both gates and analogous to Equation (11). Equation (17) is the first draft of the state before the gate $\mathbf{z}_t$ is applied and resembles Equation (12). Equation (18) is the final draft of the state after $\mathbf{z}_t$ is applied and resembles Equation (13).

The recurrent neural network is optimized using gradient descent, where the derivative of the loss function with respect to the parameters is calculated via the chain rule/reverse mode differentiation. The gradient descent optimizer algorithm we use is Adam (Kingma and Ba, 2014), which shares similarities with a quasi-Newton approach. See Appendix E.7 for more information.

E.5 The Rectified Linear Unit

A non-linearity used in our architecture, but not in the gated recurrent unit layers is the rectified linear unit (ReLU) (Agarap, 2018). The rectified linear

unit is defined as:

$$ReLU(x) = max(0, x) \tag{20}$$

The ReLU is the identity operation with a floor of zero much like the payoff of a call option. Despite being almost the identity map, this non-linearity applied in a large enough neural network can approximate any function (Hornik, Stinchcombe and White, 1989).

E.6 Skip Connections and Batch Norm

Skip connections (He et al., 2015) allow the input to skip the operation in a given layer. The input is then just added onto the output of the skipped layer, forming the final output of the layer. This allows the layer being skipped to learn a difference between the "correct" output and input, instead of learning a transformation. Additionally, if the model is overfitting, the neural network can learn the identity map easily. Skip connections are used when the input and the output are the same dimension which allows each input to correspond to one output. Because our network does not have this property, we learn a linear matrix that converts to the input to the output dimension. All the skip connections are linear operations and have no activation or batch norm, which differs from the pair of dense layers at the beginning of the network which have both batch norm and rectified linear unit activations.

Batch normalizing (Ioffe and Szegedy, 2015) is used to prevent drift of output through a deep neural network. Changes to parameters in the early layers will cause an out-sized effect on the output values for the later layers. Batch norm fixes this problem by normalizing the output to look like a standard normal distribution after the output of each layer. Thus the effect of changes in parameters will not greatly effect the magnitude of output vector as between each

output the data is re-normalized to have a mean of 0 and a standard deviation on 1.

E.7 Adam Optimizer

Adam combines momentum (Polyak, 1964)–a technique that uses recent history to smooth out swings orthonormal to the objective direction–with RMSprop (Tieleman and Hinton, 2012)–a technique used to adjust step size based on gradient volatility.

Traditional gradient descent hill climbing updates the parameters with a single equation:

$$\theta_t = \theta_{t-1} - \lambda * \nabla_\theta L_\theta(x, y) \tag{21}$$

Here $\nabla_\theta L_\theta(x, y)$ denotes taking the gradient of the loss with respect to θ, the parameters of the model. For convenience, I will denote this term g_t. By subtracting the gradient multiplied by a small step size λ, this is moving the parameters theta in the direction the reduces the loss the most at θ_{t-1}

If we wanted to use information from the second derivative to inform optimization, we can use Newton-Raphson instead:

$$\theta_t = \theta_{t-1} - H_t^{-1} * g_t \tag{22}$$

This uses the Hessian to determine an optimal step size based on steepness in the loss function. Typically, this approach is not used in deep learning as deep learning models typically have a large number of parameters and calculating the Hessian has a quadratic cost in the number of parameters and inverting also has a super-linear cost. However, there are quasi-newton methods that attempt to approximate the Hessian to determine the step size without the high

computational cost. Adam is similar to these methods. The equations that define Adam are as follows:

$$\nu_t = \beta_1 * \nu_{t-1} - (1 - \beta_1) g_t \tag{23}$$

$$s_t = \beta_2 * s_{t-1} - (1 - \beta_2) * g_t^2 \tag{24}$$

$$\delta\theta_t = -\eta \frac{\nu_t}{\sqrt{s_t + \epsilon}} * g_t \tag{25}$$

$$\theta_{t+1} = \theta_t + \delta\theta_t \tag{26}$$

The first equation is a moving average of the gradient. This "momentum" term is used because often in training the direction of the gradient would move nearly perpendicular to the direction towards the optimum. Gradient descent would spend a lot of time zig-zaging while only making slow progress towards an optimum (see Figure XVI). Taking a moving average of previous gradients preserves the principle direction while the orthogonal directions cancel each other out.

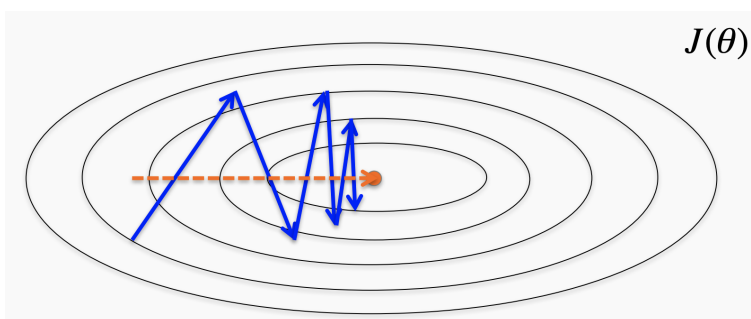

Figure XVI: Momentum

Likewise, the s_t equation is a moving average approximation for the Hessian. The approximate Hessian is used for adjusting the step size of the algorithm based on the curvature of the loss function at a given point. β_1 and β_2 are

hyperparameters that determine the smoothness of the moving average. Again the resulting update term is applied to the previous values of the parameters. This approach is empirically shown to lead to more stable optimization and even better optima than simpler gradient descent approaches for large networks.

F Additional Forecasting Information

F.1 Information Content Regressions

We regress true GDP on a varying collection of forecasts to test for statistically significant contribution of a given forecast like our gated recurrent unit model. An interpretation of significant coefficients would be that the given forecast method is adding statistically significant information when pooled with the other regressions.

Here is the median GRU forecast compared to the SPF on the baseline test set ranging from 1-quarter ahead forecasts to 5-quarters ahead:

TABLE III

	Real GDP Growth				
	(1-Qtr)	(2-Qtrs)	(3-Qtrs)	(4-Qtrs)	(5-Qtrs)
GRU	−0.195	−0.653	0.277	−0.072	0.505
	(0.541)	(0.462)	(0.414)	(0.471)	(0.517)
SPF	1.210***	2.064***	3.050***	2.957***	1.023
	(0.249)	(0.350)	(0.619)	(0.765)	(1.002)
Constant	−0.339	−1.635*	−6.370***	−5.592***	−2.161
	(0.952)	(0.953)	(1.557)	(1.970)	(2.824)
N	46	46	46	46	46
R^2	0.484	0.474	0.404	0.270	0.049
Adjusted R^2	0.460	0.449	0.377	0.236	0.005
Resid. Std. Error (df = 43)	1.880	1.898	2.020	2.237	2.553
F Statistic (df = 2; 43)	20.157***	19.367***	14.587***	7.944***	1.103

Notes: ***Significant at the 1% level.
**Significant at the 5% level.
*Significant at the 10% level.

The following table contains regressions comparing the information content of all the forecasting models (the GRU and baselines) and excluding the SPF:

TABLE IV

	Real GDP Growth				
	(1-Qtr)	(2-Qtrs)	(3-Qtrs)	(4-Qtrs)	(5-Qtrs)
GRU	1.489*	1.300	0.810	0.613	0.794
	(0.757)	(0.837)	(0.495)	(0.562)	(0.732)
DSGE	0.918*	0.137	0.168	0.055	−0.209
	(0.514)	(0.631	(0.649)	(0.675)	(0.741)
AR2	−1.139*	−1.998*	−0.645	−0.619	−0.466
	(0.640)	(1.125)	(1.768)	(1.108)	(1.922)
Factor	0.572***	0.710*	0.837	0.548*	0.440
	(0.176)	(0.388)	(0.542)	(0.325)	(0.461)
Constant	−2.655	3.156	−0.250	1.075	0.945
	(1.913)	(3.290)	(6.192)	(4.656)	(6.938)
N	46	46	46	46	46
R^2	0.444	0.158	0.120	0.080	0.057
Adjusted R^2	0.390	0.076	0.034	−0.010	−0.035
Resid. Std. Error (df = 41)	1.998	2.460	2.514	2.571	2.603
F Statistic (df = 4; 41)	8.198***	1.924	1.400	0.889	0.619

Notes: ***Significant at the 1% level.
**Significant at the 5% level.
*Significant at the 10% level.

This final table performs the same regression but has the SPF, GRU, and all baseline models:

TABLE V

	Real GDP Growth				
	(1-Qtr)	(2-Qtrs)	(3-Qtrs)	(4-Qtrs)	(5-Qtrs)
GRU	0.979	−0.332	0.277	−0.050	0.607
	(0.697)	(0.743)	(0.426)	(0.530)	(0.776)
SPF	1.140***	2.004***	3.112***	2.907***	0.903
	(0.345)	(0.404)	(0.698)	(0.818)	(1.189)
DSGE	0.661	−0.101	−0.211	−0.336	−0.301
	(0.468)	(0.504)	(0.544)	(0.606)	(0.755)
AR2	−1.408**	−0.575	0.915	−0.816	−0.829
	(0.580)	(0.941)	(1.505)	(0.979)	(1.990)
Factor	0.153	0.075	0.073	0.306	0.394
	(0.203)	(0.334)	(0.480)	(0.295)	(0.467)
Constant	−1.443	−0.270	−9.119	−2.398	0.635
	(1.755)	(2.710)	(5.498)	(4.225)	(6.986)
N	46	46	46	46	46
R^2	0.564	0.479	0.412	0.301	0.070
Adjusted R^2	0.509	0.414	0.339	0.213	−0.046
Residual Std. Error (df = 40)	1.793	1.959	2.081	2.270	2.617
F Statistic (df = 5; 40)	10.331***	7.356***	5.606***	3.438**	0.606

Notes: ***Significant at the 1% level.
**Significant at the 5% level.
*Significant at the 10% level.

F.2 Bias and Variance in the Forecasting Models

The following table contains the mean bias as well as the standard error and an indication of significance of that bias. The standard errors of the residual double as the (square-root) of the variance estimate. For the gated recurrent unit we use the median forecast:

TABLE VI

	Forecast Bias				
	(1-Qtr)	(2-Qtrs)	(3-Qtrs)	(4-Qtrs)	(5-Qtrs)
GRU	0.459*	0.480*	0.506*	0.620*	0.644**
	(0.343)	(0.369)	(0.365)	(0.379)	(0.375)
SPF	0.331	0.600**	0.723**	0.804**	0.901***
	(0.274)	(0.302)	(0.335)	(0.347)	(0.372)
DSGE	1.459***	1.513***	1.300***	1.058***	0.827***
	(0.354)	(0.378)	(0.544)	(0.386)	(0.385)
AR2	0.937***	1.317***	1.636***	1.795***	1.780***
	(0.351)	(0.381)	(0.380)	(0.383)	(0.384)
Factor	0.432*	0.163	0.459	0.533*	0.699**
	(0.328)	(0.449)	(0.367)	(0.390)	(0.414)

Notes: ***Significant at the 1% level.
**Significant at the 5% level.
*Significant at the 10% level.

Significance indicates that the forecasting model is biased in a statistical sense. Including the SPF, the above table suggests that the gated recurrent unit derives much of its outperformance due to less biased forecasts, only slightly more biased than a factor model. The factor model, though, has significant variance as illustrated on the graphs in Appendix B or gleaned from the standard deviations in the above table.

G Selected Additional Information

G.1 Reduced Form Models: US Data vs. World Data

TABLE VII: GRU Root Mean Squared Error

The performance of the best, mean, and median forecasts for the performance of gated recurrent units on the test set 2009-Q1 to 2020-Q1 (lower is better)

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
AR(2)					
US Data	2.53	2.88	3.03	3.14	3.13
World Data	2.57	2.62	2.67	2.72**	2.72**
VAR(1)					
US Data	2.69	3.02	3.14	3.11	3.20
World Data	2.46	2.60*	2.56**	2.65**	2.64**
VAR(4)					
US Data	2.99	3.03	3.10	3.08	3.08
World Data	2.37	2.52	2.56**	2.62**	2.63*

* Significance indicates outperformance of world data models over US data models

G.2 Smets-Wouters Model: US Data vs. World Data

TABLE VIII: GRU Root Mean Squared Error

The performance of the best, mean, and median forecasts for the performance of gated recurrent units on the test set 2009-Q1 to 2020-Q1 (lower is better)

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
Smets-Wouters DSGE Max Like					
US Data	3.83	4.36	4.49	4.50	4.2
World Data	3.05***	3.28***	3.28***	3.22**	3.16*
Out-Of-Sample Data	2.59***	2.75***	2.89**	3.04	3.18
1995+	2.77***	3.09**	3.22*	3.26	3.28
Smets-Wouters DSGE Bayesian					
US Data	2.79	2.95	2.89	2.80	2.71

* Significance indicates outperformance of world data models over US data models

G.3 Recurrent Neural Network Robustness Checks

For our RNN model, we found we could improve forecasting performance by taking the mean prediction of many models estimated by stochastic gradient descent. The ensembling improves performance slightly, but later graphs will show it also improves model stability and variance. Bolded entries indicate outperformance over all economic models.

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
RNN with World Data					
Best Model	2.4	2.5	**2.5**	**2.6**	**2.6**
Mean Model	2.3	2.5	**2.5**	**2.6**	**2.6**
Median Model	2.3	2.5	**2.5**	**2.6**	**2.6**

We provide a Monte Carlo simulation (Table IX), estimating our RNN model at each time horizon 100 times. At every horizon, the average root mean squared error of our simulated models indicates competitive, if not outperformance, against baseline models. Interestingly, it seems like the best performing model

on validation data, when tested on the test data, often performs worse than the average performance over all the models. This is something that should be investigated further, but based on this phenomenon, we recommend that practitioners take a simple mean or median forecast across many different models instead of picking one model.

TABLE IX: Baseline GRU Monte Carlo Simulation over Initializations

The mean and standard deviation of the performance of gated recurrent units on the test set 2009-Q1 to 2020-Q1

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
Mean RMSE	2.4	2.6	2.5	**2.6**	**2.6***
Std Dev RMSE	0.06	0.06	0.05	0.06	0.06

A common criticism of deep learning attempts at forecasting is that the models are unreliable, but due to high variance, one can p-hack a model that performs well. The Monte Carlo simulation in Table IX, anticipates this critique. The standard deviation of our RMSE is 0.06 which suggests that all our models have a similar performance on the test set when optimization is complete. This cannot resolve all issues, as the Monte Carlo result only deals with numerical instability. The model could still fit this particular data window or architecture choice, due to chance. In order to respond to those critiques, we also provide robustness checks across different architectures and data periods.

One test we preformed was to replace the gated recurrent units with a long short-term memory (LSTM) layer (Hochreiter and Schmidhuber, 1997)–another type of recurrent neural network. We use the same test data as the main result (USA 2009-Q1 to 2020-Q1) as well as the same data as inputs. The long short-term memory neural network models in Table X are analogous to the gated recurrent unit neural networks models in the table in Section VI.C. Mean RMSE and Std Dev RMSE, correspond to the entries in the table below. The baseline

performances are still the same as the test set has not changed.

TABLE X: Baseline LSTM Monte Carlo Simulations

The performance of the best, mean, and median forecasts as well as the mean and standard deviation of the long short-term memory networks on the test set 2009-Q1 to 2020-Q1 (lower is better)

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
Best RMSE	2.4	2.6	2.6	**2.6**	**2.6***
RMSE of Mean	2.4	2.6	2.5	**2.6**	**2.6***
RMSE of Median	2.4	2.5	2.5	**2.6**	**2.6***
Mean RMSE	2.4	2.6	2.5	**2.6**	**2.6***
Std Dev RMSE	0.05	0.07	0.05	0.06	0.06

The long short-term memory networks outperform the baseline models along essentially the same time horizons. Performance is also competitive, but consistently a little worse, than the gated recurrent unit over all time frames. The LSTM has similar standard deviation of root mean squared error, suggesting that the two models consistently find a similar optimum when it comes to forecasting. Again, taking a model average through the mean or median forecast results in small but consistent root mean squared error performance improvements.

Additionally, we re-estimated the model with slightly different test set from 2009-Q4 to 2019-Q4 inclusive as opposed to 2008-Q4 to 2020-Q1, comparing the benchmark economic models to our original gated recurrent unit (Table XI). The reason we use this horizon is that it contains no recessions. Since the highly flexible neural network will have an advantage forecasting periods with a significant departure from a more linear-friendly period of expansion, removing the recessions will hamstring our model compared to the more linear model baselines. Our gated recurrent units were completely re-estimated as we additionally included 2009-Q1 to 2009-Q3 in the validation set. Performance would improve if we left those (recession) time steps out of the validation set as

the test set contains no recessions, however, this decision cannot be rationalized from the point-of-view of an out-of-sample forecaster. Although this version of our model does not outperform the best baseline models along any horizon, taking into account performance over all horizons, we think our median and mean models are better than the US AR(2), VAR(1), and the factor model on this test set, while performing slightly worse than the DSGE model and the world AR(2). This supports our hypothesis that the main outperformance of our model was in highly nonlinear domains like recessions and other regime changes. However, using the cross sectional data reduced the tendency for the models to be biased upwards and was a contributor to the GRU's outperformance over models trained only on US data.

TABLE XI: Expansion Root Mean Squared Error

The performance of the best, mean, and median forecasts for the performance of gated recurrent units on the test set 2009-Q4 to 2019-Q4

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
VAR(1)					
US data	2.3	2.6	2.9	3.0	3.0
World data	2.1	2.2	2.2	2.2	2.2
AR(2)					
US data	1.7	1.7	1.8	1.9	1.9
World data	**1.6**	1.6	1.6	**1.5**	**1.5***
Smets Wouters DSGE					
US data	1.8	1.8	1.7	1.6	1.5*
Factor					
US data	1.6	**1.6**	**1.6**	1.9	2.1
GRU*					
Best	1.8	2.3	2.0	2.0	1.9
Mean Forecast	1.7	1.7	1.7	1.7	1.7
Median Forecast	1.7	1.7	1.7	1.7	1.7
SPF Median	1.4	1.5	1.5	1.5	1.5

*All GRU models use entire world data cross section

This provides supplementary evidence that the outperformance of our neural

network is not due to either over-fitting the test set or over-fitting the architecture choice. Additionally, we ran Monte Carlo simulations (Table XII) which show that given 100 random initialization and optimization routines over all 5 horizons, the model still consistently achieves low root mean squared error and has a low standard deviation–demonstrating stability and reproducability.

TABLE XII: Expansion GRU Monte Carlo Simulations

The mean and standard deviation of the performance of gated recurrent units on the test set 2009-Q4 to 2019-Q4

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
Mean RMSE	1.8	1.8	1.8	1.8	1.7
Std Dev RMSE	0.18	0.18	0.21	0.11	0.08

We also provide regressions analyzing the information content of each forecasting model by regressing true GDP on the forecast models. As none of the models except the SPF have consistent statistically significant information above and beyond the other models, we have placed that analysis in Appendix F.1. Additionally, we decompose RMSE into bias and variance of each model in Appendix F.2. In the appendix, we show that the gated recurrent unit seems to have lower bias than all models, except for the factor models. In terms of variance, our model outperforms all models, with the exception of the SPF.

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